Water Quality Classification and Contamination Risk Prediction

Project report for review · Data science project

Water-quality screening: the data is cleaned, explored and split; model training and the contamination risk score are the next stage.

1. Aim

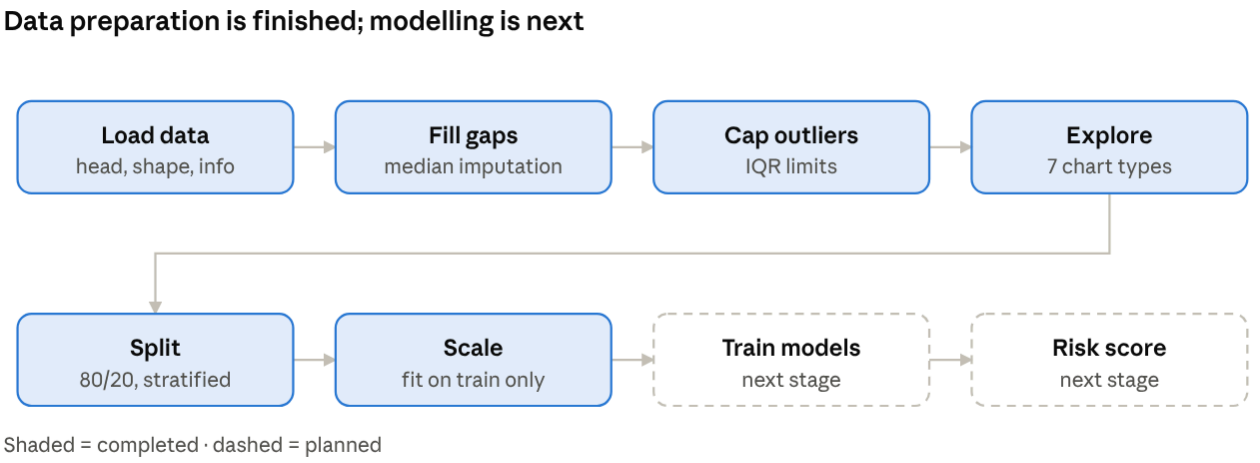
Predict from nine routine measurements whether a water sample is safe to drink (*potable*), then turn the model's probability into a Low / Medium / High contamination risk. A quick screening tool like this helps labs decide which samples to test first.

2. Data

Each row of `water_potability.csv` is one water sample: nine measurements (features) and one answer, `Potability` (1 = safe, 0 = not safe). All values are numeric, so no encoding was needed.

Feature	Meaning
ph	How acidic or alkaline the water is
Hardness	Dissolved calcium and magnesium
Solids	Total dissolved solids
Chloramines	Disinfectant used in water supplies
Sulfate	Dissolved salt; affects taste and digestion
Conductivity	How well the water carries electricity
Organic_carbon	Amount of organic matter
Trihalomethanes	Disinfection by-products; a health concern at high levels
Turbidity	How cloudy the water is

3. Workflow at a glance



project workflow · 8 steps

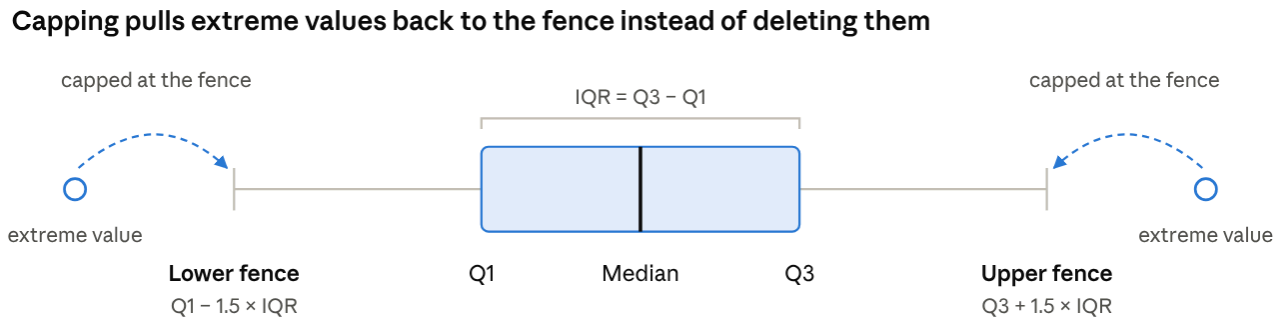
Shaded boxes are finished; dashed boxes are planned.

4. Data cleaning

- **Missing values:** ph, Sulfate and Trihalomethanes had gaps. Each was filled with its column median, which stays steady when a few readings are extreme. No rows were dropped.
- **Duplicates:** checked with `duplicated().sum()` so the model is never tested on rows it has already seen.
- **Summary statistics:** `describe()` gave the count, mean, spread and range of every feature.

5. Outliers

An outlier is a value far from the rest. I flagged values beyond 1.5 × IQR from the middle 50% of the data, capped them at that fence instead of deleting rows, and re-ran the check to confirm.



IQR capping · illustration, not project data

Capping is a design choice: an extreme reading can be a real contamination signal, so I plan to compare models with and without it.

6. Exploratory analysis

Chart	What it shows
Histogram of pH	How values are spread
Boxplot of pH	Median, spread and leftover outliers
Count plot of Potability	Whether safe and unsafe classes are balanced
Scatter: pH vs Hardness	Whether two features move together
Boxplot: pH vs Potability	Whether safe and unsafe water differ on pH
Bar chart of averages by Potability	Average pH, Hardness and Turbidity per class
Correlation heatmap	How strongly every pair of features is related

7. Preparing for modelling

- **Inputs and target:** the nine measurements form X; `Potability` is y.
- **Split:** 80% training, 20% testing. `stratify=y` keeps the safe/unsafe proportions equal in both parts, and `random_state=42` makes the split reproducible.
- **Scaling:** `StandardScaler` puts features on one scale (mean 0, standard deviation 1), since solids run into the thousands while pH is about 0 to 14. It is fitted on training data only, so no test information leaks in.

8. Next steps

- **Models:** Logistic Regression baseline, KNN, SVM, Decision Tree, Random Forest, Gradient Boosting.
- **Metrics:** accuracy, precision, recall, F1, confusion matrix and ROC-AUC, with extra attention to unsafe water wrongly called safe.
- **Risk score:** the predicted probability of "not potable", grouped into Low / Medium / High, with cut-offs chosen from validation results.
- **Refinement:** cross-validation, hyperparameter tuning and feature importance.

9. Limitations

- It is a screening aid, not a replacement for laboratory testing.
- Medians and IQR limits were computed before the split, a mild form of data leakage; a scikit-learn pipeline would fix this.
- Capping may hide genuine warning signals.
- Results depend on how the samples were collected and labelled.

10. Conclusion

The raw dataset is now clean, explored and split with a leak-free scaling step. Next comes training and comparing models, then converting their predictions into a practical contamination risk score.